

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE CODE: ITA0603

COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

CO3: Bayes Theorem – Concept Learning – Maximum Likelihood – Minimum Description Length Principle – Bayes Optimal Classifier – Gibbs Algorithm – Naïve Bayes Classifier – Bayesian Belief Network – EM Algorithm – Probability Learning – Sample Complexity – Finite and Infinite Hypothesis Spaces – Mistake Bound Model, (BL 6)

Assessment Tool Weightage: 10%

QUESTIONS:50

Total Marks:25

Annexure A

Multiple Choice Questions (MCQ)

Code of Conduct:

*I D. Sindhu (192424197) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

D. Sindhu

Signature of the student:

Annexure A

Short Answer Test

1. Bayes' Theorem is a fundamental concept in probability theory. Which of the following best describes its primary purpose in updating prior beliefs when new evidence is observed in a probabilistic model?
 - a) To compute only prior probabilities
 - b) To update prior beliefs using new evidence
 - c) To eliminate irrelevant features
 - d) To maximize classification accuracy
2. In the Naïve Bayes classifier, a strong assumption is made about the relationship between features. Which of the following statements correctly describes this assumption in practical machine learning applications?
 - a) Features are completely dependent on each other
 - b) Features are conditionally independent given the class
 - c) Features must be continuous
 - d) Features are always correlated
3. Maximum Likelihood Estimation is widely used for parameter estimation in probabilistic models. Which of the following options correctly explains the objective of MLE when fitting a model to observed data?
 - a) Maximize prior probability
 - b) Minimize hypothesis space
 - c) Maximize likelihood of observed data
 - d) Reduce computational complexity
4. The Minimum Description Length principle is often used in model selection. Which of the following best explains how MDL balances model complexity and goodness of fit in learning algorithms?
 - a) Prefers highly complex models always
 - b) Prefers simplest model that explains data well
 - c) Ignores data noise completely
 - d) Focuses only on training accuracy
5. The Bayes Optimal Classifier is considered theoretically ideal in classification tasks. Which of the following statements best describes how it achieves the minimum possible classification error?
 - a) By selecting random hypotheses
 - b) By averaging predictions over all hypotheses
 - c) By using only prior probabilities
 - d) By minimizing dataset size
6. The Gibbs Algorithm is an important concept in Bayesian learning. Which of the following options best explains how it selects a hypothesis from the hypothesis space during classification?
 - a) Selects hypothesis randomly from consistent set
 - b) Selects hypothesis with minimum loss only
 - c) Selects fixed predefined hypothesis

d) Ignores hypothesis space

7. Naïve Bayes classifiers are often used in text classification tasks. Which of the following conditions ensures that the classifier performs effectively despite its simplifying assumptions?

- a) Strong dependence between features
- b) Large amount of labeled data
- c) Features being conditionally independent
- d) Absence of noise in dataset

8. Bayesian Belief Networks are graphical models that represent probabilistic relationships. Which of the following best describes the role of nodes and edges in such networks?

- a) Nodes represent variables and edges represent dependencies
- b) Nodes represent datasets and edges represent features
- c) Nodes represent outputs only
- d) Edges represent noise in data

9. The Expectation-Maximization algorithm is commonly used when dealing with incomplete data. Which of the following best explains the role of the Expectation step in this iterative algorithm?

- a) Updates model parameters
- b) Computes expected values of hidden variables
- c) Removes missing data
- d) Sorts the dataset

10. In the EM algorithm, the Maximization step updates certain parameters. Which of the following correctly describes what is updated during this step to improve model likelihood?

- a) Hidden variables only
- b) Observed data values
- c) Model parameters maximizing likelihood
- d) Training dataset size

11. Probability learning involves estimating probability distributions from data. Which of the following best explains the goal of probability learning in the context of machine learning models?

- a) To remove all uncertainty
- b) To estimate underlying probability distributions
- c) To reduce number of features
- d) To simplify model structure

12. Sample complexity is an important concept in learning theory. Which of the following best describes what sample complexity indicates about a learning algorithm's performance?

- a) Amount of computation required
- b) Number of samples needed for learning
- c) Model accuracy only
- d) Memory consumption

13. A finite hypothesis space contains a limited number of possible hypotheses. Which of the following best explains how this limitation impacts the learning process and generalization ability?

- a) Improves flexibility always
- b) Restricts possible models considered
- c) Eliminates need for data
- d) Guarantees perfect accuracy

14. In contrast to finite hypothesis spaces, infinite hypothesis spaces allow more flexibility. Which of the following best describes the characteristics of such hypothesis spaces in learning models?

- a) Limited to discrete functions
- b) Allows continuous and complex models
- c) Contains no hypotheses
- d) Simplifies computation always

15. The mistake bound model is used to evaluate learning algorithms. Which of the following best describes what is measured by the mistake bound in online learning scenarios?

- a) Training time
- b) Maximum number of mistakes made
- c) Memory usage
- d) Dataset size

16. Prior probability is a key concept in Bayesian inference. Which of the following best describes what prior probability represents before observing any new evidence?

- a) Probability after evidence
- b) Initial belief before observing data
- c) Conditional probability only
- d) Joint probability

17. Posterior probability is calculated after incorporating evidence. Which of the following best describes how posterior probability is derived using Bayes' Theorem?

- a) Using only likelihood
- b) Using prior and likelihood with evidence
- c) Ignoring prior information
- d) Using random sampling

18. Likelihood plays a central role in statistical inference. Which of the following best explains what likelihood represents in relation to observed data and hypotheses?

- a) Probability of hypothesis given data
- b) Probability of data given hypothesis
- c) Probability of prior only
- d) Probability of error

19. Naïve Bayes classifiers are widely used in real-world applications. Which of the following best describes the type of problems where this algorithm is most effective?

- a) Clustering problems
- b) Classification problems with high-dimensional data
- c) Regression problems

d) Sorting problems

20. Overfitting is a common issue in machine learning models. Which of the following best describes a situation where a model is considered to be overfitting the training data?

- a) Model performs well on unseen data
- b) Model captures noise instead of pattern
- c) Model is too simple
- d) Model ignores data

21. Underfitting occurs when a model fails to capture patterns in data. Which of the following best describes the characteristics of an underfitted model in practice?

- a) High accuracy always
- b) Fails to learn patterns from data
- c) Too complex model
- d) Overtrained model

22. The EM algorithm is iterative and requires a stopping condition. Which of the following best explains when the algorithm is considered to have converged?

- a) When data changes
- b) When parameters stabilize across iterations
- c) When iterations increase
- d) When accuracy decreases

23. Bayesian learning combines prior knowledge with observed data. Which of the following best describes how these two components interact during the learning process?

- a) Prior replaces data
- b) Data replaces prior
- c) Both are combined to update belief
- d) Neither is used

24. Concept learning involves identifying a suitable hypothesis. Which of the following best explains the main goal of concept learning in machine learning systems?

- a) To reduce computation
- b) To find hypothesis consistent with data
- c) To eliminate features
- d) To increase dataset size

25. Hypothesis space search is a key component of concept learning. Which of the following best describes how searching through hypothesis spaces helps in selecting an appropriate model?

- a) Random selection of hypothesis
- b) Exploring possible hypotheses to find best fit
- c) Ignoring data patterns
- d) Reducing dataset size

26. In Bayesian inference, normalization ensures probabilities sum to one. What role does the evidence term play in Bayes' Theorem?

- a) It acts as a scaling factor
- b) It removes prior probability
- c) It increases model complexity
- d) It selects hypothesis randomly

27. In Naïve Bayes classification, what is the primary reason for using logarithms when computing probabilities?

- a) To increase accuracy
- b) To avoid numerical underflow
- c) To remove dependencies
- d) To simplify dataset

28. Which of the following best describes conditional probability?

- a) Probability of two events occurring together
- b) Probability of an event given another event has occurred
- c) Probability of independent events
- d) Probability of random variables

29. Which factor most affects the performance of a Naïve Bayes classifier in real-world datasets?

- a) Number of classes
- b) Independence assumption violation
- c) Dataset size only
- d) Output format

30. In Bayesian networks, what ensures that the model remains computationally efficient?

- a) Large dataset
- b) Conditional independence
- c) Continuous variables
- d) High dimensionality

31. Which of the following best describes the role of prior knowledge in Bayesian learning?

- a) Eliminates need for data
- b) Guides learning before observing data
- c) Reduces accuracy
- d) Replaces hypothesis

32. In Maximum Likelihood Estimation, what happens when the dataset size increases significantly?

- a) Prior dominates
- b) Likelihood estimation becomes more accurate
- c) Model becomes random
- d) Complexity decreases

33. Which of the following is a limitation of MLE?

- a) Uses too much prior knowledge
- b) Ignores prior information
- c) Cannot handle data
- d) Always overfits

34. Which of the following best describes Bayesian estimation compared to MLE?
- a) Uses only likelihood
 - b) Uses both prior and likelihood
 - c) Ignores uncertainty
 - d) Works only for small datasets
35. What is the main advantage of Laplace smoothing in Naïve Bayes?
- a) Removes features
 - b) Handles zero probability problem
 - c) Reduces dataset size
 - d) Increases bias only
36. Which of the following best defines a hypothesis in machine learning?
- a) Final output
 - b) Assumed model for mapping inputs to outputs
 - c) Dataset
 - d) Feature vector
37. What is the purpose of hypothesis space in learning algorithms?
- a) Store data
 - b) Define all possible models
 - c) Reduce training time
 - d) Increase features
38. Which of the following best explains generalization in machine learning?
- a) Memorizing training data
 - b) Performing well on unseen data
 - c) Reducing dataset size
 - d) Increasing parameters
39. What is the effect of noisy data on learning algorithms?
- a) Improves accuracy
 - b) Leads to overfitting
 - c) Eliminates features
 - d) Reduces complexity
40. In EM algorithm, what is the main challenge during initialization?
- a) Dataset size
 - b) Choosing initial parameters
 - c) Output format
 - d) Removing features
41. Which of the following best describes convergence in iterative algorithms?
- a) Increasing error
 - b) Stabilization of parameters
 - c) Random updates
 - d) Dataset reduction
42. Which of the following is a characteristic of probabilistic models?
- a) Deterministic outputs
 - b) Handling uncertainty

- c) Ignoring data
- d) Fixed predictions

43. What is the role of likelihood in Bayesian inference?

- a) Represents prior
- b) Represents evidence
- c) Measures fit of data to hypothesis
- d) Removes uncertainty

44. Which of the following best describes overfitting?

- a) Model too simple
- b) Model fits noise in training data
- c) Model ignores data
- d) Model has low variance

45. Which of the following best describes underfitting?

- a) Model too complex
- b) Model fails to capture patterns
- c) Model memorizes data
- d) Model has high variance

46. What is the role of training data in learning algorithms?

- a) Defines output only
- b) Guides model learning
- c) Reduces complexity
- d) Eliminates hypothesis

47. Which factor improves model generalization?

- a) Increasing noise
- b) Proper regularization
- c) Reducing data
- d) Ignoring validation

48. Which of the following best explains feature selection?

- a) Removing dataset
- b) Selecting relevant features
- c) Increasing features
- d) Ignoring attributes

49. Which of the following best describes classification problems?

- a) Predict continuous values
- b) Assign labels to data
- c) Cluster data
- d) Reduce dimensions

50. Which of the following best describes regression problems?

- a) Predict discrete labels
- b) Predict continuous values
- c) Cluster data
- d) Classify data

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	20	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	15	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

ANSWERS

1. Answer: b) To update prior beliefs using new evidence

Explanation: Bayes' Theorem combines prior knowledge with new evidence to calculate the updated (posterior) probability of an event.

2. Answer: b) Features are conditionally independent given the class

Explanation: Naïve Bayes assumes that each feature is independent of the others when the class label is known, making calculations simpler.

3. Answer: c) Maximize likelihood of observed data

Explanation: Maximum Likelihood Estimation (MLE) finds the parameter values that make the observed data most likely.

4. Answer: b) Prefers simplest model that explains data well

Explanation: The Minimum Description Length (MDL) principle selects the simplest model that explains the data without unnecessary complexity.

5. Answer: b) By averaging predictions over all hypotheses

Explanation: The Bayes Optimal Classifier considers all possible hypotheses and combines their predictions to minimize classification error.

6. Answer: a) Selects hypothesis randomly from consistent set

Explanation: The Gibbs Algorithm randomly selects one hypothesis from the set of hypotheses that are consistent with the training data.

7. Answer: c) Features being conditionally independent

Explanation: Naïve Bayes performs well when the features are approximately independent given the class, even if this assumption is not perfectly true.

8. Answer: a) Nodes represent variables and edges represent dependence

Explanation: In a Bayesian Belief Network, nodes represent random variables, and edges show the probabilistic relationships between them.

9. Answer: b) Computes expected values of hidden variables

Explanation: In the Expectation (E) step, the algorithm estimates the values of hidden or missing variables using the current model parameters.

10. Answer: c) Model parameters maximizing likelihood

Explanation: In the Maximization (M) step, the model parameters are updated to maximize the likelihood of the observed data based on the estimated hidden variables.

11. Answer: b) To estimate underlying probability distributions

Explanation: Probability learning estimates the probability distribution from the given data, helping the model make accurate predictions under uncertainty.

12. Answer: b) Number of samples needed for learning

Explanation: Sample complexity indicates the minimum number of training examples required for a learning algorithm to achieve good performance.

13. Answer: b) Restricts possible models considered

Explanation: A finite hypothesis space contains a limited number of hypotheses, making learning easier but reducing flexibility.

14. Answer: b) Allows continuous and complex models

Explanation: An infinite hypothesis space contains unlimited hypotheses, allowing more flexible and complex models to represent data.

15. Answer: b) Maximum number of mistakes made

Explanation: The mistake bound measures the maximum number of prediction errors a learning algorithm makes before learning the correct concept.

16. Answer: b) Initial belief before observing data

Explanation: Prior probability represents the initial belief about an event before any new evidence or data is observed.

17. Answer: b) Using prior and likelihood with evidence

Explanation: Posterior probability is calculated by combining the prior probability and likelihood using Bayes' Theorem after considering the evidence.

18. Answer: b) Probability of data given hypothesis

Explanation: Likelihood measures how probable the observed data is, assuming a particular hypothesis or model is true.

19. Answer: b) Classification problems with high-dimensional data

Explanation: Naïve Bayes is highly effective for classification tasks involving many features, such as spam filtering and text classification.

20. Answer: b) Model captures noise instead of pattern

Explanation: Overfitting occurs when a model learns the training data too closely, including noise, resulting in poor performance on new data.

21. Answer: b) Fails to learn patterns from data

Explanation: Underfitting occurs when the model is too simple and cannot capture the important patterns in the training data.

22. Answer: b) When parameters stabilize across iterations

Explanation: The EM algorithm converges when the model parameters change very little or remain constant between iterations.

23. Answer: c) Both are combined to update belief

Explanation: Bayesian learning combines prior knowledge with observed data to calculate the updated (posterior) probability.

24. Answer: b) To find hypothesis consistent with data

Explanation: The goal of concept learning is to identify a hypothesis that correctly classifies all training examples.

25. Answer: b) Exploring possible hypotheses to find best fit

Explanation: Hypothesis space search examines different hypotheses and selects the one that best matches the training data.

26. Answer: a) It acts as a scaling factor

Explanation: In Bayes' Theorem, the evidence term normalizes the probabilities so that the posterior probabilities sum to one.

27. Answer: b) To avoid numerical underflow

Explanation: Logarithms convert very small probability values into manageable numbers, preventing numerical underflow during computation.

28. Answer: b) Probability of an event given another event has occurred

Explanation: Conditional probability measures the likelihood of an event occurring when another event is already known to have occurred.

29. Answer: b) Independence assumption violation

Explanation: If features are highly dependent on each other, the Naïve Bayes assumption is violated, which may reduce classification accuracy.

30. Answer: b) Conditional independence

Explanation: Bayesian Networks use conditional independence to simplify probability calculations, making the model more computationally efficient.

31. Answer: b) Guides learning before observing data

Explanation: Prior knowledge represents the initial belief about a problem and helps the model make better predictions before new data is observed.

32. Answer: b) Likelihood estimation becomes more accurate

Explanation: As the dataset size increases, MLE estimates become more reliable because they are based on more observed data.

33. Answer: b) Ignores prior information

Explanation: MLE uses only the observed data for parameter estimation and does not consider any prior knowledge.

34. Answer: b) Uses both prior and likelihood

Explanation: Bayesian estimation combines prior knowledge with the likelihood of the observed data to estimate model parameters.

35. Answer: b) Handles zero probability problem

Explanation: Laplace smoothing adds a small value to feature counts, preventing zero probabilities for unseen events.

36. Answer: b) Assumed model for mapping inputs to outputs

Explanation: A hypothesis is a possible model or rule that maps input data to the correct output.

37. Answer: b) Define all possible models

Explanation: The hypothesis space contains all possible hypotheses that the learning algorithm can consider.

38. Answer: b) Performing well on unseen data

Explanation: Generalization means the model can accurately predict the output for new data that was not used during training.

39. Answer: b) Leads to overfitting

Explanation: Noisy data can cause the model to learn random variations instead of the true patterns, leading to overfitting.

40. Answer: b) Choosing initial parameters

Explanation: The EM algorithm is sensitive to its starting values, and poor initialization may lead to slower convergence or local optima.

41. Answer: b) Represents uncertainty using probabilities

Explanation: Probabilistic models use probability values to represent uncertainty and make predictions even when the information is incomplete.

42. Answer: b) Hidden variables are estimated iteratively

Explanation: The EM algorithm repeatedly estimates hidden variables and updates model parameters until convergence.

43. Answer: c) It models conditional dependencies among variables

Explanation: A Bayesian Belief Network represents the probabilistic relationships and dependencies between different variables.

44. Answer: b) Features contribute independently to prediction

Explanation: In Naïve Bayes, each feature independently contributes to calculating the probability of the target class.

45. Answer: b) Increases prediction reliability with more data

Explanation: As more training data becomes available, probability estimates become more accurate, improving prediction reliability.

46. Answer: b) Bayesian estimation is more reliable with small datasets

Explanation: Bayesian estimation uses prior knowledge along with observed data, making it more reliable than MLE for small datasets.

47. Answer: b) Candidate Elimination maintains both specific and general hypotheses

Explanation: Candidate Elimination keeps the most specific (S) and most general (G) hypotheses to represent all hypotheses consistent with the training data.

48. Answer: b) Probability learning updates model as new data arrives

Explanation: Probability learning continuously updates probability estimates using new observations, improving prediction accuracy over time.

49. Answer: b) Bayesian inference updates beliefs using new evidence

Explanation: Bayesian inference combines prior knowledge with new evidence to calculate updated (posterior) probabilities.

50. Answer: b) Probabilistic learning supports decision-making under uncertainty

Explanation: Probabilistic learning helps make informed decisions by estimating the likelihood of different outcomes even when data is uncertain or incomplete.